



From Archimedes to Gödel: The Evolution of Analysis and the Birth of the Differential Equation

(Spivak 3e · Abbott 2e · Rudin 3e · Stahl 2e)

Part I – Continuity and Paradox: Greek and Classical Origins (Weeks 1–3)

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
1	Infinity and motion (Zeno's problem)	Stewart § 1.1; Spivak Ch 1 (+ preview Ch 2); Abbott §§ 1.1–1.3	Zeno fragments; Plato <i>Parmenides</i>	Aristotle <i>Physics</i> VI; Sorabji <i>Time, Creation and the Continuum</i> Ch 1–3	What assumptions make Zeno's paradox possible? How does potential vs actual infinity differ? Can a limit exist without algebra?
2	Method of exhaustion and area	Stewart §§ 1.2–1.3; Spivak Ch 2 (+ intro Ch 5 Limits); Abbott §§ 1.4–1.5; Stahl Ch 1	Archimedes <i>Quadrature of the Parabola</i>	Heath <i>Greek Mathematics I</i> ; Boyer <i>History of the Calculus</i>	How did Archimedes anticipate integration? What is proof by exhaustion? Why geometric not algebraic?
3	Logic of proof and demonstration	Stewart § 2.4; Spivak Ch 3 (Functions); Abbott §§ 1.6–1.7	Aristotle <i>Posterior Analytics</i> ; Euclid <i>Elements</i>	Cassirer <i>Substance and Function</i> ; Sorabji Ch 4–5	How did Greek demonstration differ from modern proof? Does necessity in math imply causality?

Part II – The Scientific Revolution: Algebra, Geometry & Motion (Weeks 4–10)

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
4	Mathematizing motion	Stewart §§ 2.1–2.3; Spivak Ch 3 + Ch 4 (Graphs); Abbott §§ 2.1–2.3	Galileo <i>Two New Sciences</i>	Lloyd <i>Early Greek Science</i> Ch 7–9; Koyré <i>Galileo Studies</i>	Why did Galileo say nature is written in mathematics? What let motion become law?
5	Descartes & analytic geometry	Stewart § 1.6; Spivak Ch 4 (Graphs + Appendix on Conics); Abbott §§ 2.4–2.6	Descartes <i>La Géométrie</i>	Bos <i>Redefining Geometrical Exactness</i>	How did coordinates change proof? How did Descartes merge space and number?
6	Fermat &	Stewart §§ 2.2–2.8;	Fermat (1638); Boyer <i>Calculus</i> ;		How did Fermat

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
7	tangents	Spivak Ch 9–10 (Derivatives & Differentiation); Abbott §§ 3.1–3.3; Stahl Ch 2	Wallis (1656)	Guicciardini <i>Principia</i>	anticipate derivatives? Is the derivative an object or a process?
	Newton & Leibniz — The Invention and Unification of Calculus	Stewart §§ 4.1–5.5; Spivak Ch 11 (Significance of Derivative) + Ch 13–14 (The Integral & FTC); Abbott §§ 3.4–3.8; Stahl Ch 3–4	Newton <i>De Analysisi</i> ; Leibniz (1684)	Hall <i>Philosophers at War</i> ; Jesseph <i>Leibniz vs Newton</i>	How did the FTC resolve the paradox of motion? In what sense are differentiation and integration inverses?
	Euler — Infinite Series and the Early Concept of Function	Stewart Ch 9; Spivak Ch 22–23 (Infinite Sequences & Series); Abbott §§ 4.1–4.3; Stahl Ch 5 (Euler & Trigonometric Series)**	Euler <i>Institutiones Calculi Differentialis</i> (1755)	Cohen <i>Intro to the Principia</i> ; Hankins <i>Euler and the Mechanization of Analysis</i>	How did Euler extend calculus through series? What did it mean to represent a function by a trig expansion?
9	Fourier & Lagrange — The Analytic Ideal	Stewart Ch 11 §§ 1–6; Spivak Ch 24 (Uniform Convergence & Power Series); Abbott §§ 4.4–4.5; Stahl Ch 6 (19 th -century Analysis)**	Fourier <i>Théorie Analytique de la Chaleur</i> ; Lagrange <i>Théorie des Fonctions Analytiques</i>	Grattan-Guinness <i>Development of Analysis</i> ; Poincaré <i>Science and Hypothesis</i>	How did Fourier and Lagrange transform Euler's series into a universal analytic ideal?
10	Analytic ideal and continuity	Stewart Ch 11 §§ 8–10; Spivak Ch 24 (cont.); Abbott §§ 4.4–4.5	Fourier examples	Boyer <i>History of the Calculus</i> ; Cassirer <i>Substance and Function</i>	Why did analysis become universal? Is analysis a language or a metaphysics?

Part III – Arithmetization & Rigor (Weeks 11–20)

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
11	Real numbers & completeness	Rudin Ch 1; Spivak Ch 8 + 29–30; Abbott §§ 1.8–2.2; Stahl Ch 6	Dedekind; Cantor	Landau <i>Foundations of Analysis</i> ; Ferreirós <i>Labyrinth of Thought</i> Ch 2–3; Kline Ch 36	What does it mean to construct the continuum?
12	Sequences & limits	Rudin Ch 3 §§ 1–2; Spivak Ch 5 + 22;	Cauchy; Bolzano	Grabiner <i>Origins of Cauchy's Calculus</i> ;	Why was rigor moral for

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
13	Cauchy property & completeness	Abbott §§ 2.3–2.6; Stahl Ch 7	Cauchy	Stillwell § 9.2	Cauchy?
		Rudin Ch 3 §§ 3–4; Spivak Ch 22 (cont.) + 8; Abbott §§ 2.7–2.8; Stahl Ch 8		Kline Ch 37; Ferreirós Ch 4; Lakatos §§ 1–3	What makes a sequence Cauchy?
14	Infinite series	Rudin Ch 3 §§ 5–8; Spivak Ch 23; Abbott §§ 4.1–4.3; Stahl Ch 9	Euler; Fourier	Grattan-Guinness Ch 5; Kline Ch 38	Why were divergent series dangerous?
15–16	Power series & analytic functions	Rudin Ch 8; Spivak Ch 24; Abbott §§ 4.4–4.5; Stahl Ch 9 (cont.)	Weierstrass; Riemann	Bottazzini Ch 5–6; Dauben Ch 4	How did Riemann and Weierstrass redefine function and continuity?
17	Uniform convergence	Rudin Ch 7; Spivak Ch 24 (Uniform section); Abbott §§ 4.4–4.5	Dirichlet; Weierstrass	Grattan-Guinness Ch 6; Kline Ch 40	Why uniform rather than pointwise?
18	Existence & uniqueness (ODEs & integral eqns.)	Rudin Ch 5 + 6 §§ 1–2; Spivak Ch 14 review; Abbott § 3.12 (Applications of FTC); Boyce-DiPrima Ch 9	Picard; Peano	Cassirer <i>Problem of Knowledge</i> Ch 3	How does integration clarify existence?
19	Stability & dynamics	Rudin Ch 4; (no direct Spivak); Abbott Ch 4 summary + § 8.4; Arnold Ch 1	Poincaré; Lyapunov	Smale “Differentiable Dynamical Systems”	How did Poincaré redefine solution and stability?
20	Philosophy of rigor	Review Rudin 1–3; Spivak Ch 28–30 (Fields and $\mathbb{R}$); Abbott §§ 8.3–8.5	Poincaré <i>Value of Science</i>	Cassirer Ch 4; Whitehead <i>Science and the Modern World</i> Ch 6	Did rigor separate math from experience?

Part IV – Measure, Integration & Abstraction (Weeks 21–32)

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
21–22	Measure & Integration — From Riemann to Lebesgue	Rudin Ch 6; Spivak Ch 13–14 (revisit for Riemann basis); Abbott §§ 7.1–7.3 (The Riemann Integral and Beyond); Folland Ch 1–	Lebesgue; Borel	Grattan-Guinness Ch 7; Bottazzini Ch 7; Dieudonné § 1; Cassirer <i>Problem of Knowledge</i> Ch 5	How does measure extend area? Why generalize Riemann?

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
		3; Royden Ch 4			
23–24	Hilbert Spaces & Functional Analysis — Geometry of Functions	Rudin <i>RCA</i> Ch 3–5; Abbott § 7.4 (Functions as Points in Infinite-Dimensional Spaces)	Hilbert (1906)	Weyl <i>Raum, Zeit, Materie</i> Ch 7; Courant & Robbins Ch 8; Dieudonné Ch 2	Why infinite dimension? How did Hilbert unify analysis and algebra?
25–28	Fourier Analysis & Operators — The Analytical Heart of Modern Physics	Rudin <i>RCA</i> Ch 6–8; Spivak Ch 14, 24 (revisit foundations of continuity & series); Abbott §§ 8.1–8.2 (Fourier and Operator Ideas in Analysis)	Fourier; Riesz	Bottazzini Ch 8; Kline Ch 42; Weyl <i>The Continuum</i> Ch 2; Cassirer <i>Determinism and Indeterminism</i> § 2	Why did Fourier analysis become central? Is operator theory a new metaphysics of function?
29–30	Set Theory & Foundations — The Arithmetization of the Infinite	Mendelson; Enderton; Spivak Ch 28–30 (Reals and Fields, revisited); Abbott § 1.9 + §§ 8.3–8.4 (Logic and Foundations)	Cantor; Zermelo; Russell	Ferreirós Ch 5–6; Dauben Ch 6; Corfield Ch 2	How did logic replace intuition as foundation?
31–32	Gödel & Formalism — The Limits of Systematic Knowledge	Rudin <i>RCA</i> Ch 11–12; Abbott §§ 8.3–8.5 (Limits of Formal Systems)	Gödel; Hilbert	Cassirer <i>Problem of Knowledge</i> Ch 6; Corfield Ch 3; Lakatos <i>Proofs and Refutations</i> (Postscript)	What did Gödel prove about consistency? Are there limits to mathematical reason?

Part V – ODE Expansion Module (Weeks 33–36)

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
33	Linear Systems of ODEs — From Euler to Cauchy	Boyce–DiPrima Ch 7–8; Teschl § 2; Review Spivak Ch 14 (FTC results); Abbott § 3.12 (Applications of the FTC and ODE Examples); Rudin Ch 5 (revisit)	Euler; Cauchy	Arnold <i>ODEs</i> Ch 2; Struik <i>Source Book</i>	How do matrices model coupled systems?
34	Phase Space & Eigenmethods — The Geometry of	Teschl §§ 3–4; Arnold Ch 2; Abbott § 7.4 (Functional Spaces)	Liouville; Jacobi	Smale <i>Dynamical Systems</i>	What does phase space reveal about motion?

Week	Focus	Core Texts	Primary Sources	Historical & Philosophical Readings	Reflection Questions
	Motion	revisited) Strogatz Ch 1–4;			
35	Nonlinear Dynamics & Chaos — The End of Determinism	Arnold Ch 3; Abbott § 8.2 (Fourier and Operator Framework for Dynamics)	Poincaré; Lorenz	Prigogine <i>Order Out of Chaos</i> Ch 2	What is chaos mathematically?
36	Numerical Methods & Computation — The Algorithmic Continuum	Atkinson §§ 9.1–9.3; Teschl § 5; Abbott § 8.5 (Approximation and Computation)	—	Press <i>Numerical Recipes</i> Intro	How does approximation redefine solution and truth in analysis?

Chronological and Conceptual Flow Summary

Historical Phase	Core Concepts	Spivak (3e)	Abbott (2e)	Modern Counterparts
Greek Foundations (Weeks 1–3)	Infinity, exhaustion, proof	1–3	1.1–1.7	Stewart Ch. 1–2
Scientific Revolution (Weeks 4–10)	Function, derivative, integral, series	4, 9–14, 22–24	2.1–4.5	Stahl 2–6
Rigor & Arithmetization (Weeks 11–20)	Completeness, convergence, uniformity	5, 8, 22–24, 29–30	1.8–4.5, 3.12	Rudin 1–8
Abstraction & Measure (Weeks 21–32)	Integration, functional analysis, set theory	13–14, 24, 28–30	7.1–8.5	Rudin/Folland
Modern Dynamics (Weeks 33–36)	ODEs, chaos, computation	14 (review)	3.12, 7.4, 8.2–8.5	Arnold/Strogatz

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Weekly Workflow: “Analytic–Reflective Cycle”

(Designed for the 36-week From Archimedes to Gödel curriculum)

Day	Focus	Typical Activities	Goal / Rationale
Saturday – The	Core study:	• Read Spivak , Stewart , and	Begin the week with

Day	Focus	Typical Activities	Goal / Rationale
Deep Dive	major theorems, definitions, worked examples	• Abbott sections for the week. • Take detailed notes on proofs and intuitive structure. • Solve warm-up problems from Abbott and Spivak. • Attempt Rudin problems or proofs from Spivak. • Work through at least one <i>historically significant problem</i> (e.g. Fermat tangent, Euler series). • Write concise summaries of each theorem's meaning. • Finish the week's Rudin or Stahl sections. • Check connections to ODEs or measure theory. • Summarize one key concept (e.g. completeness, uniform convergence). • Read the week's Stahl, Boyer, or Bottazzini chapter. • Take margin notes linking historical development to proofs studied earlier. • Read short excerpts from primary sources (e.g. Archimedes, Descartes, Weierstrass). • Read Cassirer, Poincaré, Whitehead, or Weyl. • Ask: what philosophical tension is this week's math addressing? • Optional: brief journaling (200 words). • Answer the week's <i>Reflection Question</i> in 1–2 paragraphs. • Summarize relationships between formalism, intuition, and history. • Revise notes to capture “big picture.” • Skim next chapters in Stewart or Abbott. • Identify upcoming definitions or problem themes. • Write 3–5 preview questions for Saturday.	long, uninterrupted focus; absorb core technical material while fresh. Move from comprehension to creation; reinforce structure through problem-solving. Anchor understanding in formal language and logic before shifting gears. Reconnect mathematical work to its origins; foster narrative understanding without heavy calculation. Transform technical insights into philosophical reflection. Synthesize conceptual and technical understanding into coherent insight. Prime your mind for the next analytic cycle; end week with curiosity, not fatigue.
Sunday – Problem Immersion	Proofs, exercises, creative exploration		
Monday – Formal Rigor & Review	Abstraction and consolidation		
Tuesday – Context & History (Light Day)	Intellectual and historical framing		
Wednesday – Philosophy & Interpretation (Light Day)	Conceptual digestion		
Thursday – Synthesis & Reflection	Integrative writing and discussion		
Friday – Preview & Reset	Light preview of next week's math		

Weekly Time Distribution (Flexible Guide)

Focus Area	Recommended Time	Purpose
Core Texts (Stewart, Spivak, Rudin, Abbott)	8–10 hrs	Foundation and mastery of formal analysis.
Problem Solving	4–6 hrs	Proof practice, examples, and review.
Historical Readings (Stahl, Boyer, Bottazzini, Grattan-Guinness)	2–3 hrs	Context and historical comprehension.
Philosophical Readings (Cassirer, Poincaré, Whitehead, Weyl, Lakatos)	2–3 hrs	Reflection on meaning, method, and foundations.
Reflection / Writing	1 hr	Weekly synthesis and conceptual integration.

Cycle Philosophy

1. **Start with concrete mastery** (Saturday–Monday): build structure, intuition, rigor.
2. **Shift midweek to reflection** (Tuesday–Friday): process ideas through history and philosophy.
3. **Return to abstraction on Saturday**: each week becomes a spiral of increasing synthesis — *mathematics* → *meaning* → *mathematics again*.